

FINDING THE AREA OF COMPOSITE FIGURES



LESSON INTRODUCTION

MA.7.GR.1.2 Solve mathematical or real-world problems involving the areas of polygons or composite figures by decomposing them into triangles or quadrilaterals.

LEARNING TARGETS

- I can decompose composite figures into triangles and quadrilaterals to find the area.

BENCHMARK COHERENCE

BUILDING ON

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

WORKING TOWARD

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

ESSENTIAL QUESTIONS

- How do you decompose composite figures into triangles and quadrilaterals to find the area?
- How do you find the area of a composite figure inscribed in a rectangle?

LESSON NARRATIVE

Students explore the similarities and differences between decomposing a composite figure to find its area and inscribing that composite figure within a rectangle to find the area in a different way. Students interact with triangles and rectangles as they work to decompose composite figures while making meaning of the purpose and intent of inscribing within a rectangle.

COMPONENT SUMMARY

4.4.1 WARM-UP (~5 MIN.)

Identify math errors in an advertisement

4.4.3 COLLABORATION (10 MIN.)

Collaboratively decomposing a composite figure on grid paper to find its area

4.4.5 YOUR TURN! (5 MIN.)

Find the area of a composite polygon

4.4.2 GUIDED INSTRUCTION (20 MIN.)

Find areas of composite figures by decomposing and by inscribing them in rectangles

4.4.4 GUIDED PRACTICE (10 MIN.)

Practice decomposing composite figures to find their areas

4.4.6 WRAP-UP (~5 MIN.)

Reflect on mastery of lesson learning targets

RESOURCES

GLOSSARY

Key Terms:

- Composite figure

Additional Terms:

- N/A

MATERIALS

- (Optional) Graph paper
- (Optional) Scissors
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- After decomposing the composite figures, students may struggle to find the lengths of sides not given in the triangles or quadrilaterals.

THINGS TO CONSIDER

- In 4.4.4 Guided Practice prompt students to create a table with the types of shapes they decompose the figures into. Creating a table (like the one in 4.4.3 Collaboration) will help students organize their work.
- Consider allowing students to use calculators.
- If time is an issue, then consider assigning 4.4.4 Guided Practice and/or 4.4.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

During 4.4.1 Warm-Up prompt students to initially think about the math fail and write their response independently. Next, prompt students to share their thinking with a partner. Partners can provide each other with feedback before the whole-class discussion.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

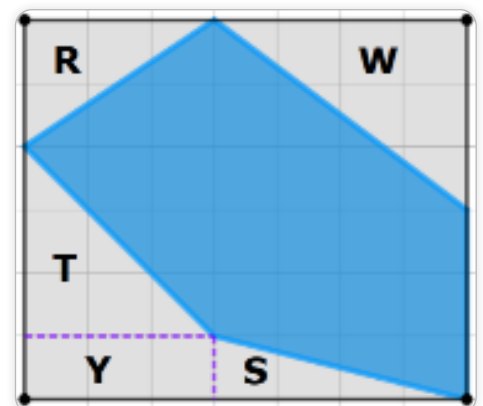
MA.K12.MTR.4.1 Discussion and Reflection

Students collaborate with a partner in 4.4.3 Collaboration as they find the areas of composite figures. Students have opportunities to discuss their thinking with their peers. Select and highlight student work that advances and deepens the development of correct and increasingly efficient methods.

DIFFERENTIATE INSTRUCTION

In 4.4.3 Collaboration question 1, students can use a graphic organizer to keep track of similarities and differences, with the potential to use different colors to show their partner's method of decomposing the figure.

In 4.4.2 Guided Instruction consider prompting students to use graph paper to draw the figure and inscribe it in a rectangle. Students can cut out each shape that is not a part of the composite shape. Then, students can calculate and record the area for each one.



4.4.1 WARM-UP: MATH FAIL

TEACHER MOVES

Encourage creativity and thinking outside of the box. Students could respond in different ways, and both be 'correct' depending on their justification.

- Students may identify a 3:1 ratio, stating that two threads should cost \$6.
- Students may believe that unit price should decrease as quantity increases to encourage people to buy more. So, if buying 2 for \$5 saves a dollar, then buying 3 could cost \$7 to save \$2.



4.4.1 Warm-Up: Math Fail

The advertisement below is for the sale of colored strands for braids to raise money for a Stop Drug Abuse campaign.

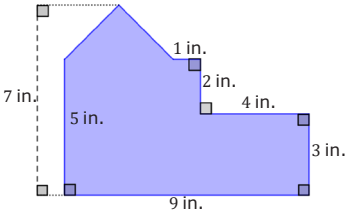


1. Explain what the math fail is in the advertisement.
Answers will vary. The math fail is that the rates are inconsistent; the 5 to 2 rate is not the same as the other two 3 to 1 rates.

4.4.2 Guided Instruction: Finding the Area of Polygons

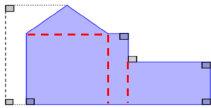
Composite figures are two- or three-dimensional figures that can be decomposed into smaller figures.

A floorplan is shown. All segment lengths are in inches (in.).



1. With your partner, decompose the floorplan into triangles and quadrilaterals.

 Answers will vary.



2. Find the area of the floorplan. Show your work.

 $A = \frac{1}{2}bh$; $A = \frac{1}{2} \cdot 4 \cdot 2$; $A = \frac{1}{2} \cdot 8$; $A = 4 \text{ in}^2$

 $A = bh$; $A = 4 \cdot 5$; $A = 20 \text{ in}^2$

 $A = bh$; $A = 1 \cdot 5$; $A = 5 \text{ in}^2$

 $A = bh$; $A = 4 \cdot 3$; $A = 12 \text{ in}^2$

 Total Area: $4 + 20 + 5 + 12 = 41 \text{ in}^2$

4.4.2 GUIDED INSTRUCTION: FINDING THE AREA OF POLYGONS

ENRICHMENT

What shapes did you decompose your figure into? How can you use the given dimensions to determine the unknown dimensions?

TEACHER MOVES

Think-Pair-Share

Prompt students to initially think about their mistakes and write their responses independently. Next, prompt students to share their thinking with a partner. Partners can provide each other with feedback before the whole-class discussion.

3. A polygon inscribed inside a rectangle is shown on the grid. Each grid line is 1 unit.

 a. Complete the table.

Polygon Inscribed in a Rectangle
Decomposition of Rectangle
Area of the large rectangle 42 unit ²
Area of R 3 unit ²
Area of S 2 unit ²
Area of T 4.5 unit ²
Area of W 6 unit ²
Area of Y 3 unit ²
Area of the blue polygon 23.5 unit ²

DIFFERENTIATE INSTRUCTION

Consider having students use graph paper to draw and cut out a 6 by 7 unit rectangle. Then, students can draw and cut out R, S, T, Y, and W. This will provide a visual and kinesthetic entry-point for calculating the areas in the decomposition of the rectangle.

4.4.2 GUIDED INSTRUCTION: FINDING THE AREA OF POLYGONS (CONTINUED)

ENRICHMENT

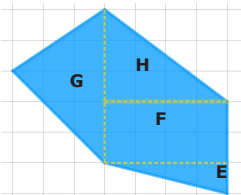
- What are the advantages of inscribing a composite figure into a rectangle to find area? What are the disadvantages?
- Which method do you prefer to use?

TEACHER MOVES

Think-Pair-Share

Prompt students to initially think about their mistakes and to write their responses independently. Next, prompt students to share their thinking with a partner. Partners can share feedback with each other before the whole-class discussion.

- b. The same polygon is shown on the grid. Each grid line is 1 unit.

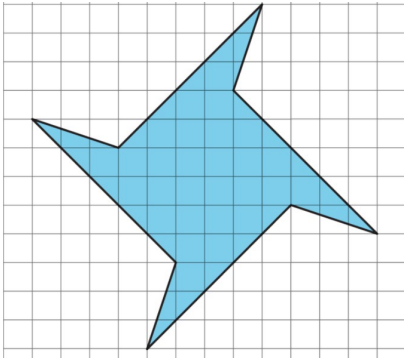
Polygon

Decomposition of Polygon
Area of E 2 unit^2
Area of F 8 unit^2
Area of G 7.5 unit^2
Area of H 6 unit^2
Area of the blue polygon 23.5 unit^2

- c. Discuss with your partner the different possible methods of finding the area of the blue polygon. List one strength of each method.
Answers will vary. One method is to subtract the area surrounding the blue polygon from the area of the large rectangle. A strength is using shapes already defined. Another method is to decompose the blue polygon into various shapes and to find the total area of the shapes. A strength is finding the total area of less shapes.

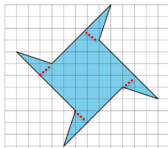


4.4.3 Collaboration: Area of a Polygon

1. A polygon is shown on grid paper. Each grid line is 1 unit.



- a. Decompose the polygon into triangles and quadrilaterals.



- b. Share your decomposition with your partner. List at least one similarity and one difference.

Answers will vary. Ours is similar because we decomposed it into the same amount of polygons. Ours is different because we decomposed it into a different amount of triangles and quadrilaterals.

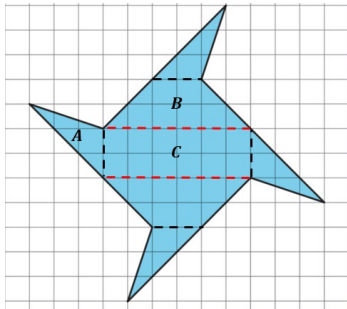
4.4.3 COLLABORATION: AREA OF A POLYGON

DIFFERENTIATE INSTRUCTION

Provide colored pencils for students to sketch on their Workbook for support in decomposing, or provide copies of the shape on the coordinate plane for students to cut.

2. Taahira decomposed the polygon into 7 pieces using 3 different shapes, as shown. Each grid line is 1 unit.

Work with your partner to find the area of each shape and the total area of the polygon.



Shape	Type of Shape	Area of the Shape	Number of Shapes	Total Area
A	triangle	3 unit ²	4	12 unit ²
B	trapezoid	8 unit ²	2	16 unit ²
C	rectangle	12 unit ²	1	12 unit ²
Area of the Polygon				40 unit ²

TEACHER MOVES

Consider having students decompose to find the total area while also inscribing within a rectangle to verify their answers. Students can work in pairs, with one student decomposing like Taahira while the partner is inscribing to verify. Students could also choose their own method with a partner before finding someone around the room who did it a different way to verify if they got the same answers.

4.4.4 GUIDED PRACTICE: DECOMPOSING COMPOSITE FIGURES TO FIND AREA

TEACHER MOVES

Encourage students to draw either:

- the decomposed triangles and quadrilaterals
- the rectangle in which the figure is inscribed

Prompt students to label the dimensions and make a table to organize the areas of the triangles and quadrilaterals.

ENRICHMENT

What method did you use to determine the area? How can you use the given dimensions to determine the unknown dimensions?

QUESTIONING

Consider challenging students who finish early to find the areas of the two figures using a different approach. Prompt students to recognize that this is a way to check their work.

4.4.4 Guided Practice: Decomposing Composite Figures to Find Area

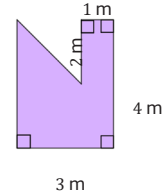
1. A polygon is shown on a grid. Each grid line is 1 unit.



Find the area of the polygon.

36 unit²

2. A polygon is shown with the given measurements in meters (m).



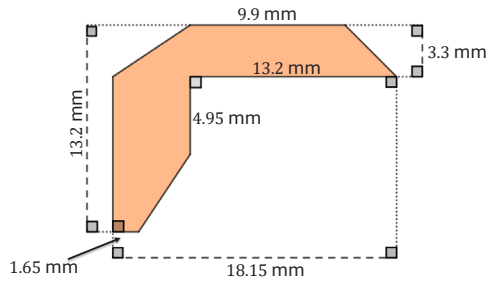
What is the area of the composite figure?

10 m²



4.4.5 Your Turn!

- Find the area of the polygon. Lengths are shown in millimeters (mm).



87.12 mm^2

4.4.5 YOUR TURN!

QUESTIONING

Consider challenging students who finish early to draw and label their own composite figures. Then, prompt students to swap figures with a partner and calculate the area of their partner's figure.



4.4.6 Wrap-Up: Reflection

- Place an **X** on the number in the emoji scale that indicates your understanding in relation to today's learning target.

Answers will vary.

I can decompose composite figures into triangles and quadrilaterals to find the area.



0 1 2 3 4 5 6 7 8 9 10

- Provide evidence to explain your self-assessment.

Answers will vary.

4.4.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider creating a checklist designed for your learning targets. Students can create an emoji for each item or learning target.

- I can decompose composite figures into triangles and quadrilaterals.
- I can label the decomposed figures with the correct measures.
- I can apply the correct formulas to find the area for each decomposed figure.
- I can correctly calculate rational numbers to find the area.